ARGUS: Adaptive Real-time Green-time Update System Using Multi-Camera YOLO Vehicle Detection and PCU-Based Weighting at a Four-Leg Intersection

Compiled by:
Nabil Akbar Aprianto
Alvaro Arta Putra
Emir Javas Kawisworo
Baihaqi Raihan Cahyadi
Banyu Danadyaksa Karuna Gusti
mira

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TABLE OF CONTENTS

TABLE OF CONTENTS 1
ABSTRACT 3
CHAPTER 1 4
Introduction 4
1.1. Background of the Study 4
1.2. Problem Statement 5
1.3. Objectives of the Research 5
1.4. Significance of the Research 5
1.4.1. Theoretical Significance 5
1.4.2. Practical & Applied Significance 6
1.5. Research Hypotheses 6
CHAPTER 2 7
Literature Review 7
2.1. Theoretical Framework 7
CHAPTER 3 9
3.1. Analisis Kebutuhan dan Karakteristik Sistem 9
3.1.1 System Requirements Analysis 9
3.1.2 System Characteristics 9
3.2. Perancangan Arsitektur dan Algoritma Sistem 9
3.2.1. Input 9
3.2.2. Proses 10
3.3. Implementasi Perangkat Lunak dan Integrasi Modul 11
3.3.1. Implementasi Perangkat lunak 11
3.3.1.1. YOLO26n-Based Vehicle Detection 11
3.3.1.2. Audio Classification and Siren Detection 11
3.3.1.3. PCU-Based Traffic Load Estimation and Signal Control 11
3.3.1.4. Graphical User Interface and Research Data Logging 11
3.3.2. Integrasi Modul 11
3.4. Experimental Setup and Testing Scenarios 11
3.4.1. Simulation Environment & Configuration 11
3.4.2. Heterogeneous Traffic Density & PCU Weighting Protocol 11
3.4.3. Dual-Modal Emergency Vehicle Preemption Protocol 11
3.4.4. Transition Mechanism & Time-Compensation Validation 11
3.4.5. Environmental Robustness & Signal Degradation Testing 11
3.5. Performance Metrics and System Evaluation 11
3.5.1. Computer Vision & Object Detection Metrics 11
3.5.2. Traffic Flow & Queue Reduction Metrics 12

3.5.3. Preemption Latency & Dual-Modal Verification Accuracy.....12
3.5.4. Signal Recovery & Transition Equilibrium.....12
CHAPTER 4..... 13
Hasil dan pembahasan..... 13
4.1. Pengujian Computer Vision.....13
4.2. Evaluasi Pengaturan Lalu Lintas.....13
4.3. Pengujian Prioritas Kendaraan Darurat.....13
4.4. Evaluasi Pemulihan Sistem.....13
4.5. Analisis Keseluruhan Sistem.....13
CHAPTER 5..... 14

ABSTRACT

Rapid urbanization in D.I. Yogyakarta has led to severe traffic congestion, exacerbated by fixed-time traffic controllers that cannot adapt to dynamic vehicular flow or account for heterogeneous traffic compositions. Furthermore, traditional systems fail to prioritize emergency vehicles efficiently, leading to critical transit delays. To address these limitations, this study proposes an Intelligent Traffic Light System integrating computer vision-based dynamic signal allocation with dual-modal emergency vehicle preemption. The proposed architecture employs YOLO26n to detect, classify, and quantify multi-class vehicle density in real time, mapping traffic volume to Passenger Car Unit (PCU) weighted densities to allocate proportional green light durations bounded by strict minimum and maximum thresholds. Simultaneously, emergency vehicle preemption is triggered through a dual-modal verification framework combining audio signal feature extraction via Mel-Frequency Cepstral Coefficients (MFCC) from siren acoustic data with visual object detection. Upon emergency vehicle detection, a time-capped override grants immediate intersection clearance, followed by a dynamic time-compensation mechanism to restore signal equilibrium across displaced approaches without causing secondary congestion spikes. Experiments conducted on a scaled intersection prototype evaluate detection accuracy, signal transition latency, and green time distribution fairness across various traffic density scenarios. Results demonstrate that the integrated PCU-weighted adaptive algorithm significantly reduces overall intersection delay and queue lengths compared to conventional fixed-time setups, while the dual-modal preemption mechanism successfully creates an uninterrupted corridor for emergency responders. This research provides a scalable, low-cost Intelligent Transportation Systems (ITS) framework suitable for municipal traffic retrofitting in high-density urban environments.

Keywords: *Adaptive Traffic Control, Computer Vision, Emergency Vehicle Preemption, Multi-Class Vehicle Weighting, Intelligent Transportation Systems.*

CHAPTER 1

Introduction

1.1. Background of the Study

Rapid urbanization and escalating vehicular density pose critical challenges to modern urban mobility, directly undermining Sustainable Development Goal 11 (SDG 11: Sustainable Cities and Communities) by straining physical road infrastructure. Inefficient traffic signal control further exacerbates idling times and greenhouse gas emissions, countering SDG 13 (Climate Action). Moreover, conventional pre-timed traffic signals fail to prioritize emergency response vehicles, causing crucial transit delays that compromise public safety and emergency preparedness under SDG 3 (Good Health and Well-being).

In developing urban centers such as the Special Region of Yogyakarta, Indonesia, rapid vehicular growth exceeded 3.5 million registered units in 2025 on a largely static road network. Yogyakarta's traffic landscape is characterized by high heterogeneity, where unsegregated lanes are shared by motorcycles, private passenger cars, public transit, and heavy freight. Conventional pre-timed traffic controllers are fundamentally incapable of responding to fluctuating traffic volumes, causing clear approaches to receive unnecessary green time while congested lanes experience prolonged queuing. Furthermore, standard controllers rely on uniform vehicle counting, neglecting spatial occupancy variations between vehicle classes (e.g., motorcycles vs. heavy trucks).

These limitations are intensified by the absence of intelligent emergency vehicle preemption, leaving first responders trapped in traffic queues. Existing adaptive traffic solutions often rely on single-modal detection systems (vision-only or audio-only), which remain highly vulnerable to visual occlusions, adverse weather, or ambient urban noise.

To address these challenges, this study proposes an Intelligent Transportation System (ITS) framework named ARGUS, replacing rigid fixed-time controllers with a computer vision-driven adaptive signal allocation model powered by a fine-tuned YOLO26n object detection network. The system applies dynamic Passenger Car Unit (PCU) weighting across eight vehicle classes to compute spatial occupancy, allocating bounded green durations to prevent lane starvation. Additionally, a dual-modal verification framework combining visual recognition with YAMNet audio embeddings and complementary Mel-Frequency Cepstral Coefficients (MFCC) acoustic feature extraction enables reliable emergency preemption with a time-compensation mechanism. Finally, this research evaluates detection accuracy, signal transition latency, and green time distribution fairness across density scenarios via a scaled intersection prototype.

1.2. Problem Statement

1. How can a dynamic Passenger Car Unit (PCU)-based vehicle weighting model be formulated for heterogeneous traffic to allocate green time, both under normal conditions and during emergency vehicle preemption?
2. How should a signal transition mechanism be designed following the end of a preemption period, so that the return to normal adaptive mode does not generate a new traffic build-up at the intersection's approaches?
3. How effective is the integrated model of emergency vehicle priority and PCU-based weighting in reducing emergency vehicle travel time, and what is its impact on the delay time of non-emergency vehicles?

1.3. Objectives of the Research

This study aims to design and develop a computer vision-based traffic light control system capable of detecting the presence of vehicles in real time using a webcam, enabling the system to determine lane priorities adaptively and fairly for road users and vehicles. This prioritization is implemented through a time-capped override algorithm, a mechanism that grants absolute green-light priority to emergency vehicles while still providing time compensation to other lanes that were temporarily disadvantaged. In addition, this study aims to develop and test a miniature intersection prototype to evaluate the accuracy of vehicle detection, system response time, and the fairness of green-light time distribution among lanes. Therefore, this study is expected to reduce the response time for emergency vehicles without excessively compromising the smooth flow of traffic on other lanes.

1.4. Significance of the Research

1.4.1. Theoretical Significance

1. **Advancement in Heterogeneous Traffic Modeling:** Contributes to the field of Intelligent Transportation Systems (ITS) by establishing an algorithmic model that accounts for multi-class, non-homogeneous traffic flow (varying vehicle ratios like motorcycles vs. cars) rather than relying on uniform vehicle count logic
2. **Integration of Vision-Based Computer Vision and Real-Time Preemption Logic:** Enriches the literature on computer vision applications by combining real-time spatial object detection (YOLO) with dynamic priority override algorithms for emergency vehicles.
3. **Framework for Low-Cost Computational Traffic Control Systems:** Provides a theoretical benchmark for executing real-time vision-based traffic algorithms on budget-friendly hardware without relying on high-cost computational infrastructure or complex machine learning models.

1.4.2. Practical & Applied Significance

1. **Reduction of Urban Traffic Congestion and Idling Time:** Minimizes unnecessary vehicle waiting times by dynamically allocating green light durations based on actual road volume weight (V_i), improving overall intersection throughput.
2. **Cost-Effective Infrastructure Retrofitting for Local Municipalities:** Offers a scalable, budget-friendly solution for city traffic authorities to upgrade existing traffic light infrastructure without requiring expensive sensor overhauls.
3. **Optimization of Emergency Response Times:** Drastically reduces transit delays for emergency vehicles (ambulances, police cars, fire trucks) by automatically creating an uninterrupted “Green Wave” corridor at intersections.

1.5. Research Hypotheses

1. **H0 (Null Hypothesis):**
   The computer vision model does not possess a significant or reliable detection accuracy and response speed to trigger timely traffic light signal changes.
2. **Ha (Alternative Hypothesis):**
   The computer vision model possesses a significant and reliable level of detection accuracy and response speed to trigger timely traffic light signal changes.

CHAPTER 2

Literature Review

2.1. Theoretical Framework

2.1.1. Traffic Flow Dynamics and Heterogeneous Characteristics

Traffic flow at signalized intersections depends dynamically on volume (V), speed (v), and density (k) [5]. In urban areas like D.I. Yogyakarta, rapid vehicular growth causes severe congestion because road capacity remains imbalanced [6]. Local traffic is mixed and heavily dominated by motorcycles. Thus, simple unweighted vehicle counts cannot accurately measure true road occupancy [1].

To bring diverse vehicle classes into a single metric, we apply the Passenger Car Unit (PCU) concept [7]. In this study, discrete PCU weights are defined as 0.25 (or 0.33) for motorcycles, 1.00 for light vehicles, 2.00 for medium heavy vehicles/buses, and 2.00 for heavy vehicles/trucks. These discrete PCU values simplify standard MKJI ranges to reduce computational overhead during real-time edge computing and dynamic signal calculation. We formulate the weighted traffic demand (Q_i) on approach i as follows:

$$Q_i = \sum_k (w_k \cdot n_{i,k})$$

where n_{i,k} is the count of class-k vehicles on approach i, and w_k is the PCU weight [7]. This formula lets the control algorithm allocate green times dynamically based on real-time spatial occupancy [3], [4].

Table 2.1: Passenger Car Unit (PCU) equivalent factors for heterogeneous traffic

2.1.2.

Vehicle Classification	PCU Value (w_k)	Reference Standard
Motorcycle (MC)	0.25 / 0.33	MKJI / Sugiarto et al. [7] (Simplified)
Light Vehicle / Passenger Car (LV)	1.00	Baseline standard [7]
Medium Heavy Vehicle / Bus (MHV)	2.00	MKJI / Sugiarto et al. [7] (Simplified)
Heavy Vehicle / Large Truck (HV)	2.00	MKJI / Sugiarto et al. [7] (Simplified)

2.1.3. Computer Vision

Computer vision helps Intelligent Transportation Systems (ITS) extract and analyze visual data automatically from roadside cameras [5]. For real-time vehicle detection and counting, the model implements YOLO26n (architecture fine-tuned for 8 custom vehicle classes), featuring an anchor-free detection head and a C2f module. This setup speeds up feature extraction while keeping

accuracy high, as shown in Figure 2.1 [5]. When deployed on edge devices at intersections, this architecture detects, classifies, and counts multi-class vehicles effectively [5]. In addition, modern computer vision pipelines help handle poor lighting, occlusions, and heavy traffic density [5].

2.1.4. Acoustic Signals

Acoustic signals travel as pressure waves defined by frequency (Hz) and amplitude, providing vital supplementary data for identifying emergency vehicles visually occluded by traffic queuing [3]. Siren acoustics produce distinct spectral signatures compared to ambient urban noise. The primary audio subsystem utilizes YAMNet audio embeddings combined with a lightweight classifier (Logistic Regression C=100 / SVM RBF) to extract 1024-dimensional feature vectors, while Mel-Frequency Cepstral Coefficients (MFCC) via librosa and spectral analysis serve as complementary diagnostic tools for acoustic signal characterization and confidence thresholding [8]. Integrating acoustic classification with visual object detection forms a robust dual-modal preemption framework, minimizing false triggers while ensuring rapid intersection clearance [3], [8].

2.1.5. Control Algorithm

A traffic control algorithm determines how a signal controller converts real-time inputs into dynamic timing decisions. Fixed-time controllers follow rigid timing plans regardless of demand [5], whereas adaptive controllers dynamically adjust green durations based on volume.

Proportional allocation and bounded green-time algorithms formulated by Osman et al. [3] and Abbas et al. [2] calculate signal durations directly from vehicle density. Teslyuk et al. [4] defined bounded green-time calculations where only detections above confidence threshold (c_threshold) are counted:

$$N_i = \sum (c_i \geq c_{\text{threshold}})$$

The resulting green duration is constrained within minimum and maximum limits to prevent lane starvation:

$$T = \max(T_{\min}, \min(T_{\max}, T_{\text{raw}}))$$

Position of this study: Existing adaptive algorithms focus primarily on raw counts or fuzzy logic (e.g., Rahayu [1]), rarely integrating multi-class PCU weighting with dual-modal emergency preemption. This study expands proportional green-time allocation using PCU-weighted density:

$$Q_i = \sum_k (w_k \cdot n_{i,k})$$

where n_{i,k} is class-k count and w_k is PCU weight [7]. Final green duration (G_i) is:

$$G_i = \max\left(G_{\min}, \min\left(G_{\max}, \frac{Q_i}{\sum_j Q_j} \cdot G_{\text{total}}\right)\right)$$

Upon siren verification, a time-capped override grants immediate green priority, followed by dynamic time-compensation to restore traffic equilibrium smoothly.

CHAPTER 3

3.1. Analisis Kebutuhan dan Karakteristik Sistem

System Requirements Analysis and Characteristics ARGUS (Adaptive Real-time Green-time Update System) is a computer vision based system that is developed to optimize traffic signal timing based on real-time traffic conditions. Previous studies have explored computer vision based vehicle detection and adaptive traffic signal control [1]–[3]. ARGUS use YOLO26n for vehicle detection using multiple CCTV feeds and webcams, Passenger Car Unit (PCU) for weighting and to estimate traffic load [7], and an emergency vehicle prioritization mechanism based on siren detection.

3.1.1 System Requirements Analysis

ARGUS requires integration with Yogyakarta CCTV feeds as visual data sources and webcams for experimental testing, alongside the YOLO26n model to detect and count vehicles across multiple approaches of a four-leg intersection. The detection results are processed using PCU weighting to estimate traffic load and compare conditions across approaches [2], [7]. This analysis serves as a basis for adjusting green light durations while maintaining appropriate safety constraints. Furthermore, the system requires a siren detection mechanism to prioritize emergency vehicles, as well as real-time processing capabilities, operational reliability, software compatibility, and an interface that presents traffic information clearly.

3.1.2 System Characteristics

ARGUS is characterized by computer vision based vehicle detection using YOLO26n and multi camera monitoring to achieved traffic information from different intersection [1], [2]. The system applies PCU weighting to estimate traffic load based on vehicle categories, including motorcycles, passenger cars, buses, and trucks [7]. The result estimates support adaptive green-light timing adjustments according to observed traffic conditions [2], [4]. Additionally, ARGUS incorporates an emergency vehicle prioritization mechanism based on siren detection and uses Yogyakarta CCTV feeds and webcams as visual data sources.

3.2. Perancangan Arsitektur dan Algoritma Sistem

3.2.1. Input

Sistem menerima dua jenis input, yaitu visual dan audio. Input visual berasal dari empat sumber kamera *CCTV* yang mewakili lajur Utara, Timur, Selatan, dan Barat. Sistem memproses video menggunakan *Computer Vision* dengan

model YOLO26n untuk mendeteksi dan mengklasifikasikan delapan jenis kendaraan, termasuk ambulan, mobil pemadam kebakaran, dan mobil polisi.

Input audio berasal dari file MP3 yang dipilih pengguna melalui GUI. Sistem menganalisis karakteristik frekuensi suara menggunakan pustaka librosa untuk menghasilkan nilai confidence sebagai indikasi keberadaan sirene.

Kualitas kedua input mempengaruhi hasil analisis. Deteksi visual bergantung pada kualitas video dan ketepatan model, sedangkan analisis audio menggunakan pendekatan heuristik sehingga nilai *confidence* belum dapat dianggap sebagai probabilitas pasti bahwa suara tersebut merupakan sirene.

3.2.2. Proses

Dalam pengolahan data kami menggunakan data yang kami peroleh dari computer vision untuk menentukan bobot setiap lampu merah sehingga waktu untuk setiap lampu merah bisa ditentukan. Hal tersebut dapat dilakukan dengan cara menghitung total kendaraan berdasarkan bobot masing masing kendaraan, yang kemudian data tersebut dimasukan kedalam rumus seperti yang terlampir di bab 2. Selain untuk menentukan waktu lampu merah, data yang diperoleh digunakan untuk mendeteksi kendaraan darurat agar sistem dapat melakukan sistem darurat yaitu memprioritaskan jalur yang terdeteksi secara suara dan computer vision.

3.2.3. Output

Antarmuka sistem pengendali lampu merah dirancang untuk memudahkan operator dalam memantau kondisi lalu lintas dan mengamati proses pengaturan lampu lalu lintas secara adaptif. Antarmuka ini terdiri atas beberapa bagian, yaitu panel kamera yang menampilkan kondisi empat lajur, meliputi Utara, Timur, Selatan, dan Barat, beserta hasil deteksi kendaraan; panel informasi kendaraan yang menampilkan jumlah kendaraan berdasarkan jenisnya, yaitu sepeda motor, mobil, bus, dan truk; serta panel kontrol lampu lalu lintas yang menampilkan status lampu merah, kuning, dan hijau beserta durasi waktunya. Selain itu, terdapat panel pengaturan untuk memilih mode pengoperasian dan mengendalikan simulasi, serta panel Audio Confidence yang menampilkan nilai kepercayaan deteksi suara, seperti sirine. Antarmuka ini digunakan untuk memantau hasil deteksi kendaraan dan proses simulasi pengaturan lampu lalu lintas berdasarkan kepadatan kendaraan. Pada tahap pengembangan ini, pengendalian lampu lalu lintas fisik melalui perangkat IoT belum terintegrasi ke dalam sistem.

3.3. Implementasi Perangkat Lunak dan Integrasi Modul

3.3.1. Implementasi Perangkat lunak

3.3.1.1. YOLO26n Vehicle Detection

3.3.1.2. YAMNet and MFCC-Based Audio Classification and Siren Detection

3.3.1.3. PCU-Based Traffic Load Estimation and Signal Control

3.3.1.4. Graphical User Interface and Research Data Logging

3.3.2. Integrasi Modul

3.4. Experimental Setup and Testing Scenarios

3.4.1. Simulation Environment & Configuration

We tested ARGUS on a 4-leg miniature intersection prototype. Four webcam feeds sent live video to an edge PC running Python and YOLO26n [5]. High-quality MP3 siren files provided audio inputs via a custom GUI, with the audio subsystem utilizing YAMNet audio embeddings coupled with a lightweight classifier (Logistic Regression C=100 / SVM RBF) to extract 1024-dimensional feature vectors, alongside librosa-based MFCC extraction and spectral analysis as complementary diagnostic tools [8].

3.4.2. Heterogeneous Traffic Density & PCU Weighting Protocol

Tests covered low, medium, and heavy rush-hour traffic [5, 6]. We mixed motorcycles, cars, buses, and trucks [7]. Green times ($G_i adjusted$) adjusted dynamically based on PCU spatial load (Q_i), kept within G_min and G_max limits to avoid starving empty lanes [2, 4].

3.4.3. Dual-Modal Emergency Vehicle Preemption Protocol

Simulations evaluated responses to ambulances, fire trucks, and police cars [3]. We tested visual occlusions where big buses blocked emergency vehicles. This proved the dual-modal setup combining YOLO26n visual detection with YAMNet embeddings and complementary MFCC acoustic analysis outperforms single-modal systems under line-of-sight loss [3, 8].

3.4.4. Transition Mechanism & Time-Compensation Validation

We tracked signal recovery after emergency overrides ended [3]. A dynamic time-compensation algorithm added green seconds to delayed approaches. This restored normal traffic flow without triggering secondary jams [2, 3].

3.4.5. Environmental Robustness & Signal Degradation Testing

We benchmarked YOLO26n accuracy across day, night, and rainy lighting [5]. YAMNet audio classification and MFCC siren checks were tested against urban background noise to measure Signal-to-Noise Ratio (SNR) limits and avoid false alerts [3, 8].

3.5. Performance Metrics and System Evaluation

3.5.1. Computer Vision & Object Detection Metrics

YOLO26n accuracy was tested across target vehicle classes using Precision (P), Recall (R), and mean Average Precision (mAP) [5]:

$$P = {}^{TP} / {}_{(TP + FP)}, \quad R = {}^{TP} / {}_{(TP + FN)}$$

$$mAP = {}^{1}/_{N} \sum_{i=1}^{N} AP_i$$

Here, TP, FP, and FN mark true positives, false positives, and false negatives above confidence threshold (c_threshold >= 0.7) [4, 5]. Evaluation tracked mAP50 and mAP50-95 under low light and heavy occlusions [5].

3.5.2. Traffic Flow & Queue Reduction Metrics

We compared PCU dynamic allocation against fixed-time signals using Average Intersection Delay ($(\bar{D})$) [1, 7]:

$$\bar{D} = \left(\sum_{i=1}^{4} \sum_{j=1}^{N_i} d_{i,j}\right) / \left(\sum_{i=1}^{4} N_i\right)$$

where d_{i,j} is the waiting delay of vehicle j on approach i [2, 3]. We also tracked queue length reduction ($(\Delta L_{\text{queue}})$) and green-time fairness across all approaches [2, 4].

3.5.3. Preemption Latency & Dual-Modal Verification Accuracy

Preemption efficiency was measured via trigger latency ((L_{preempt})), tracking the time from initial detection to green light activation [3]. Dual-modal reliability was assessed by comparing false positive rates (FPR) against single-modal baselines under visual obstruction and street noise [3, 8].

3.5.4. Signal Recovery & Transition Equilibrium

We measured the recovery cycle count ((C_{rec})) needed to return to normal adaptive equilibrium [3]. Delay recovery ratios on non-emergency lanes verified that no secondary traffic spikes formed after overrides ended [2, 3].

CHAPTER 4

Hasil dan pembahasan

4.1. Pengujian Computer Vision
Hasil deteksi kendaraan dan metrik akurasi YOLO26n

4.2. Evaluasi Pengaturan Lalu Lintas
PCU, pembagian waktu hijau, waktu tunggu, dan antrean

4.3. Pengujian Prioritas Kendaraan Darurat
Deteksi visual, audio, konfirmasi ganda, dan latency

4.4. Evaluasi Pemulihan Sistem
Kondisi lalu lintas setelah prioritas darurat

4.5. Analisis Keseluruhan Sistem
Kelebihan, kekurangan, dan keterbatasan Lampiu

CHAPTER 5

Conclusion (Arial, 12, Bold)

(Arial, 12 pt) Summarize the key conclusions drawn from the study evaluating YOLO26n and the dual-modal YAMNet/MFCC acoustic emergency vehicle preemption system. Do not restate the abstract or the entire set of results. Emphasize the contribution and potential applications of the findings.

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